

Inspection Platform Overview

Flight planning, capture and analytics

Flight planning

Each asset gets a stored plan: waypoints, altitudes, camera angles and overlap. The plan is version controlled, so cycle twelve is flown exactly as cycle one and the imagery is directly comparable.

Capture

Visual capture at 45 megapixels with 80 percent forward overlap. Thermal capture on electrical assets. Every frame carries position, altitude and heading metadata.

Defect detection

Models trained on 41,000 annotated defect instances across corrosion, cracking, insulator damage, conductor strand breakage and vegetation encroachment. Precision 88.6 percent on the internal test set at the operating threshold.

Change detection

The value of the second cycle is the difference from the first. Findings are matched across cycles by position so an owner sees which defects grew, which are stable and which are new.

Human review

Every report is reviewed by an analyst before it is released. No finding reaches a client unread. This is why report turnaround is 72 hours rather than 2, and it is also why no client has disputed a finding.